

GCP Service Account Key Rotation Automation

Automated, Secure, and Auditable Cloud Credential Lifecycle Management

DEVELOPED BY THE DEVOPS TEAM @ MOVATE

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What Is This System?

Every application running on Google Cloud uses a **service account key** — a digital credential that proves the app's identity and grants it access to resources. Like any password, these keys must be **regularly changed** to stay secure. When managed manually across dozens of accounts and projects, this process is slow, error-prone, and often missed — creating security gaps and the risk of sudden application outages.

This automation **handles the entire process for you** — scanning, rotating, validating, and reporting — every week, without any manual effort.



Fully Automated

Runs on a weekly schedule with zero manual intervention required.



Zero Downtime

Old credentials stay active during transition — no service disruption.



Multi-Project Coverage

Covers all GCP projects across the organisation in a single run.



Full Audit Trail

Every rotation is logged and reported — supporting compliance and governance.

Bottom line: This system ensures that no service account credential ever becomes a security liability — eliminating manual work, preventing outages, and giving leadership complete visibility after every run.

Overview & How It Works

1

Scan

Reviews all service account credentials across every configured GCP project.

2

Detect

Flags credentials expiring within 14 days for proactive rotation.

3

Rotate

Generates new secure credentials and stores them in a protected vault.

4

Report

Sends a full status report via email to all stakeholders.

WHAT YOU RECEIVE AFTER EACH RUN

- ▶ **HTML Email Report** — colour-coded summary of all service accounts: which were rotated, which are healthy, and which need attention.
- ▶ **Excel Attachment** — full detail for every credential: project, account name, rotation date, status, and storage location.
- ▶ **Delivered automatically** — no action needed to receive the report. It arrives after every weekly run.

YOUR 3-STEP GUIDE (AFTER RECEIVING THE REPORT)

- ▶ **Review the report** — identify any service accounts with a "Rotated" status. These are the ones that need new credentials distributed.
- ▶ **Coordinate with your team** — the DevOps team retrieves the new credentials from the secure vault and passes them to the relevant application team.
- ▶ **Confirm & close** — the application team updates their system with the new credentials and confirms everything is working. The old key is then retired.

KEY OUTCOMES AT A GLANCE

- Weekly automated runs
- No manual key rotation
- Email + Excel reports
- Zero service disruption
- All projects covered
- Compliance ready

Problems Solved, Advantages & Limitations

✗ BEFORE — THE PROBLEM

- ✗ Keys rotated manually — slow, inconsistent, and frequently missed
- ✗ No central visibility — no audit trail across projects
- ✗ Expired keys cause application outages and failed API calls
- ✗ Long-lived credentials increase the security attack surface

✓ AFTER — THE SOLUTION

- ✓ Fully automated rotation on a weekly schedule — zero manual effort
- ✓ Complete audit trail — every rotation logged and reported
- ✓ Proactive detection 14 days ahead — rotated before expiry
- ✓ Regular rotation enforced across all projects — security posture improved

KEY ADVANTAGES



Bank-Grade Security

Credentials are generated offline and stored in a secure, access-controlled vault. Nothing sensitive is ever exposed over the network.



Zero Downtime Rotation

Old and new credentials coexist during the handover window — applications keep running without interruption.



Scales Effortlessly

Adding a new GCP project to the rotation scope requires changing a single configuration value — no code changes.



Compliance Ready

Supports SOC 2, ISO 27001, and other frameworks that require regular credential rotation with documented evidence.

KNOWN LIMITATIONS

Manual distribution to applications required

App teams must update credentials & restart services

Old keys stay active briefly until manually retired

Report-based handover — not a push deployment

Architecture Diagram

High-level view of how all components connect and communicate

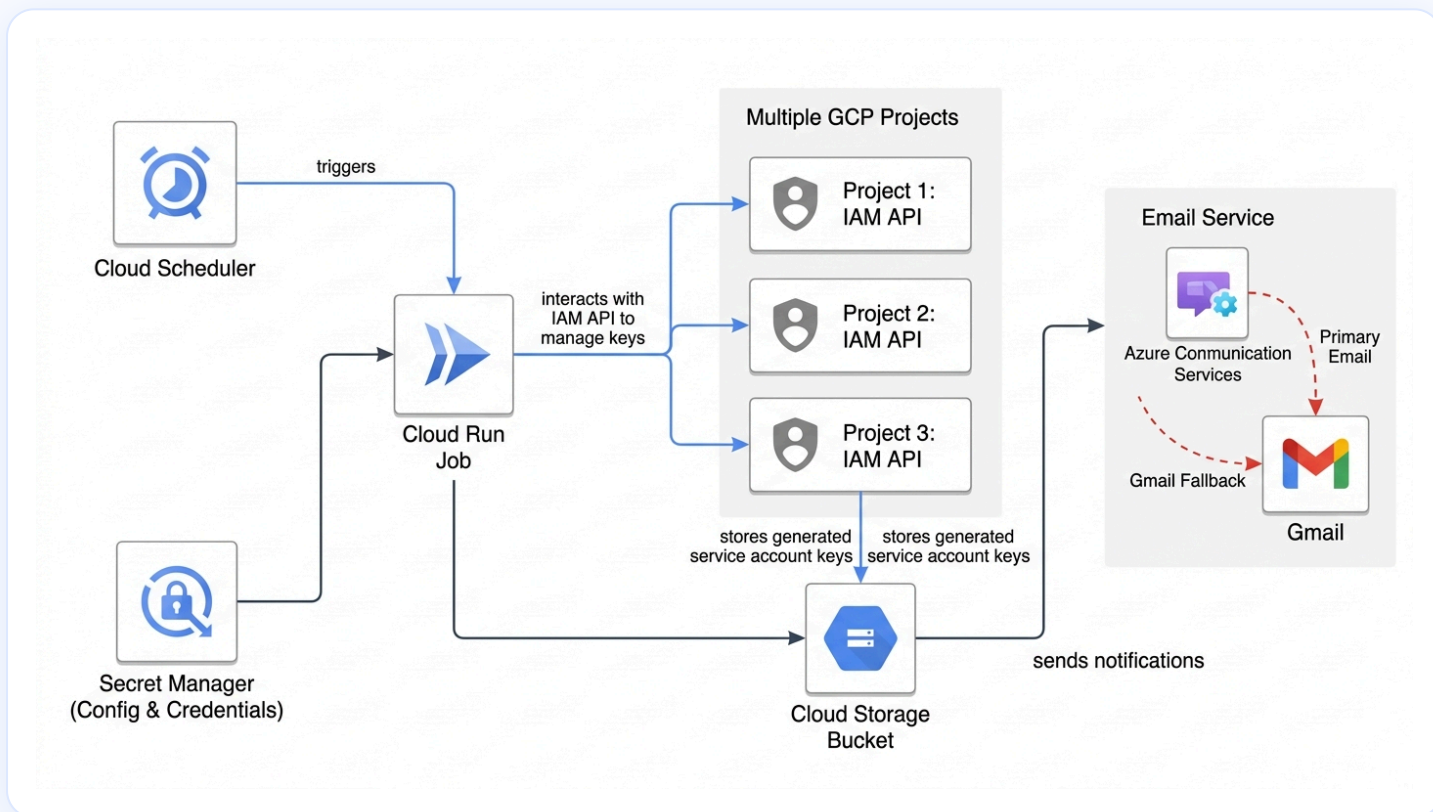


Figure 1 — GCP Service Account Key Rotation: High-Level Architecture

Output: Email Report & Excel Attachment

Sent automatically to all stakeholders after every weekly run

After each run, two reports are delivered automatically — a colour-coded **HTML email** for a quick overview, and a detailed **Excel attachment** for deeper review, record-keeping, and compliance evidence.

HTML EMAIL REPORT

CLOUD OPS

GCP Service Account Key Report

Automated key expiry scan and rotation summary

GCP Service Account Key Report [Scan Report] — 2026-04-17

Hi Team,

Scan Report

The attached Excel workbook contains the full results for all projects. Keys marked **Expiring** are within the alert threshold and require attention.

KEYS CHECKED	EXPIRING (&LE;THRESHOLD)	SAFE / OK	ERRORS
3	1	2	0

Key Expiry Scan Results

All USER, MANAGED keys across configured projects

SERVICE ACCOUNT	PROJECT	EXPIRY DATE	DAYS REMAINING	STATUS
billing-report-sa@finops-billing-central-prod.iam.gserviceaccount.com	finops-billing-central-prod	2026-06-24	68	OK
test-service-account-gemini-a@gemini-api-test-492112.iam.gserviceaccount.com	gemini-api-test	2026-04-23	6	Expiring
514946790027-compute@developer.gserviceaccount.com	vm-info-collector-server	2026-05-10	23	OK

Full details in attachment: [sa_key_scan_report_2026-04-17.xlsx](#)

Auto

Notice: This is an automatically generated email. For questions or concerns, please contact the DevOps / Cloud Ops team.

This message was sent automatically. Please do not reply.

Colour-coded status summary with key metrics and per-account details

EXCEL ATTACHMENT

Project Name	Project ID	Service Account	Expiry Date	Days Remaining	Status	Storage Location	Rotation Timestamp	Error	Key Name
finops-billing-central-prod	finops-billing-central-prod	billing-report-sa@finops-billing-central-prod-2026-06-24.iam.gserviceaccount.com	2026-06-24	68	OK				
gemini-api-test	gemini-api-test-492112	test-service-account-gemini-a@gemini-api-test-492112.iam.gserviceaccount.com	2026-04-23	6	Expiring				
vm-info-collector-server	vm-info-collector-server	514946790027-compute@developer.gserviceaccount.com	2026-05-10	23	OK				

Full 10-column report with filters, rotation timestamps, and storage locations

Thank You

For questions, feedback, or a live demo of the system,
please reach out to the DevOps team below.

CONTACT THE TEAM

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